Biostatistics Project

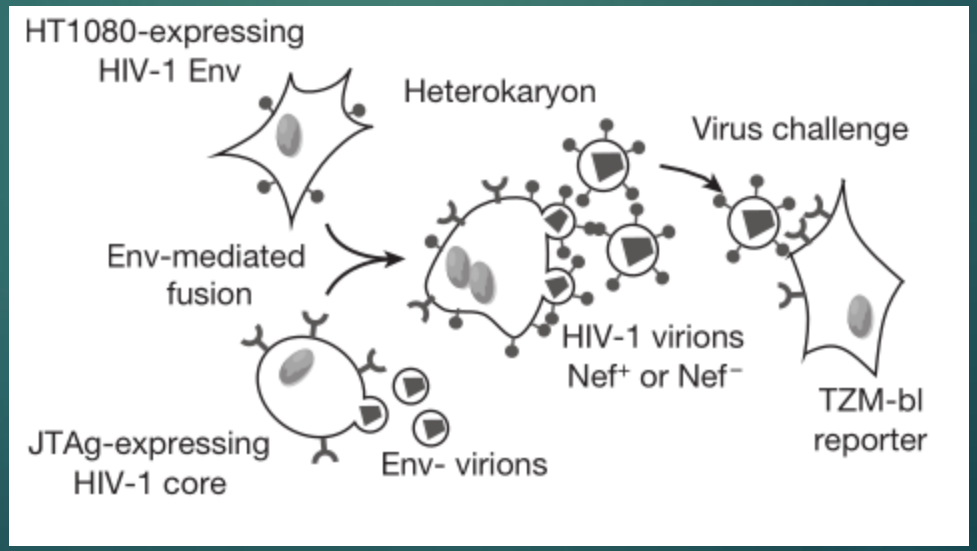
HIV-1 Nef & SERINC5 Analysis

REPLICATION AND ANALYSIS OF ROSA ET AL. 2015

(DHRUVI SHETH, 22294)

Background

- HIV-1 Nef increases infectivity of viral particles
- SERINC5 identified as restriction factor
- Goal: replicate + extend findings using RNA-seq data



Data

- 15 cell lines (RNA-seq HTSeq counts)
- Infectivity ratio (Nef+/Nef-)
- Ensembl gene expression → RPM → log2 transform

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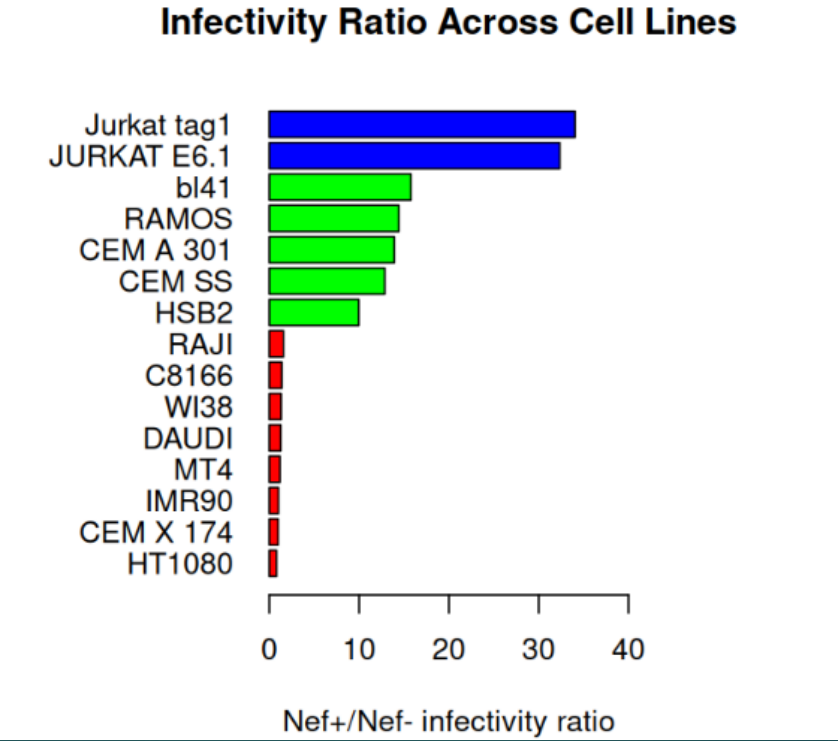
Infectivity data loaded.
          SRR infectivity_ratio   cell_line
1  SRR2166626              0.79      HT1080
2  SRR2166634              0.98   CEM X 174
3  SRR2166632              1.02       IMR90
4  SRR2166624              1.19         MT4
5  SRR2166629              1.27       DAUDI
6  SRR2166635              1.31        WI38
7  SRR2166630              1.38       C8166
8  SRR2166627              1.57        RAJI
9  SRR2166625              9.96        HSB2
10 SRR2166628             12.85      CEM SS
11 SRR2166633             13.92   CEM A 301
12 SRR2166631             14.44       RAMOS
13 SRR2166637             15.78        bl41
14 SRR2166636             32.35 JURKAT E6.1
15 SRR2166638             34.04 Jurkat tag1
```

Methods Overview

- RNA-seq preprocessing (count matrix, RPM, log2)
- Correlation analysis (Pearson, Spearman, Kendall)
- FDR correction (Benjamini-Hochberg)
- Group comparison (Wilcoxon test)
- PCA for global structure

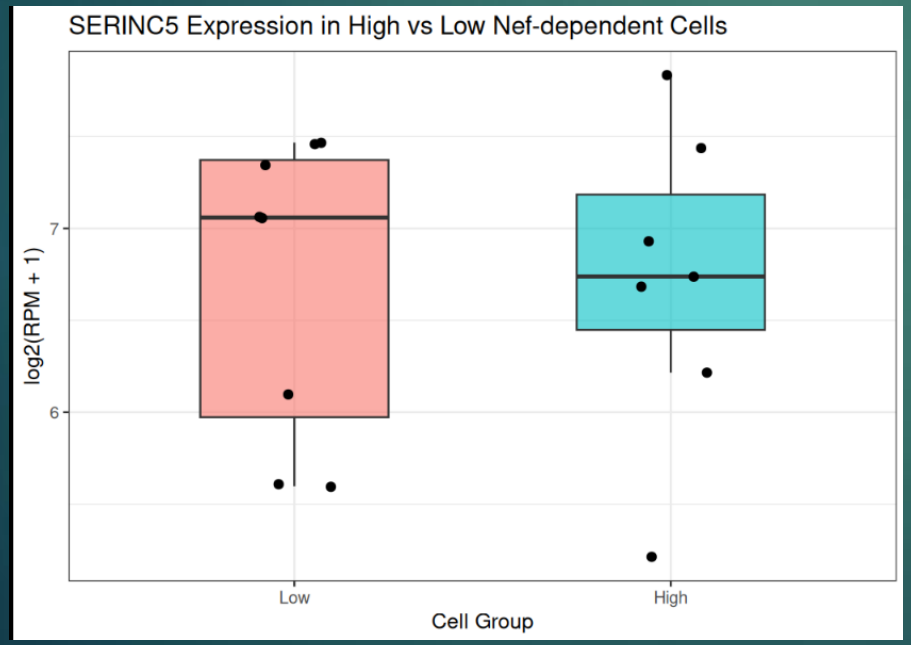
Replication of Rosa et al.

- • SERINC5 vs Infectivity correlation
- • SERINC3 vs Infectivity correlation



H1: SERINC5 in High vs Low Nef Cells

- • Wilcoxon rank-sum test
- • Permutation test validation
- RESULTS: INSERT p-value + boxplot

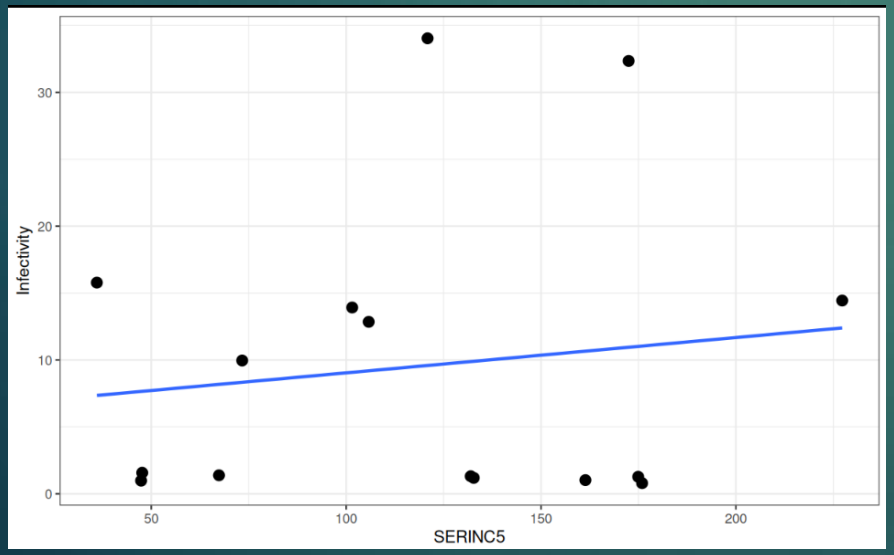


Wilcoxon rank sum exact test data: SERINC5 by group W = 30, p-value = 0.4333
alternative hypothesis: true location shift is greater than 0

Observed difference: 0.01061049
Permutation p-value: 0.498

H2: SERINC5 vs Infectivity

- • Pearson correlation (linear)
- • Spearman correlation (rank-based)
- • Kendall correlation (robust check)
- RESULTS:



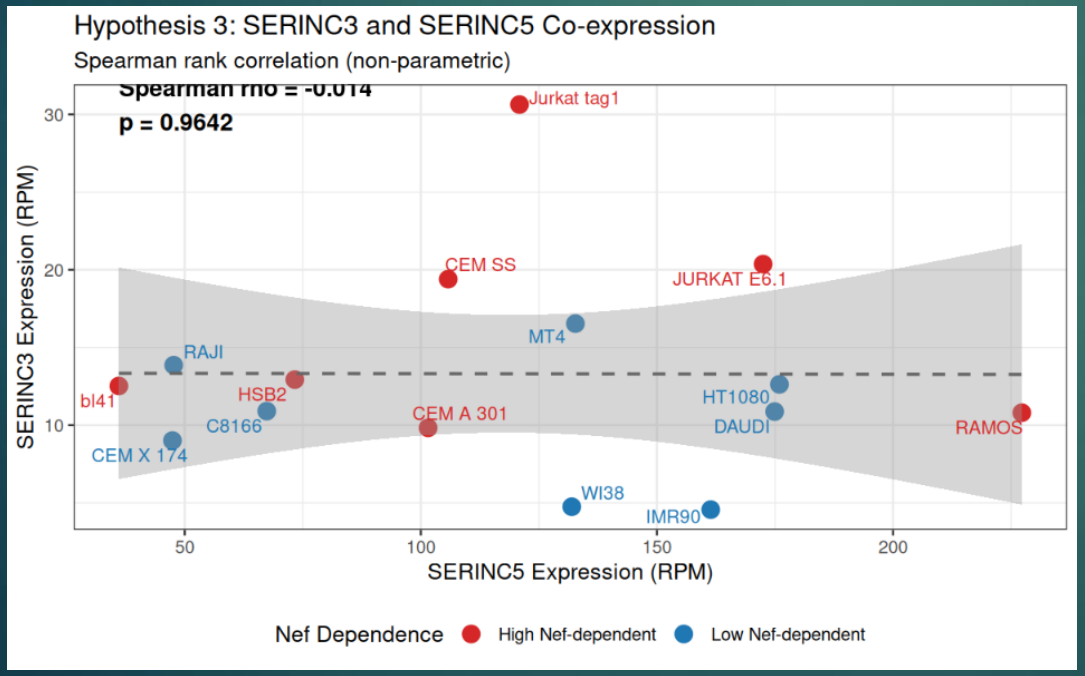
Spearman correlation:
rho = -0.1285714
p-value = 0.6482018

Kendall correlation:
tau = -0.1047619
p-value = 0.6264945

Pearson correlation:
r = 0.1336321
p-value = 0.6349339

H3: SERINC3–SERINC5 Co-expression

- Spearman correlation analysis
- Tested transcriptional co-regulation



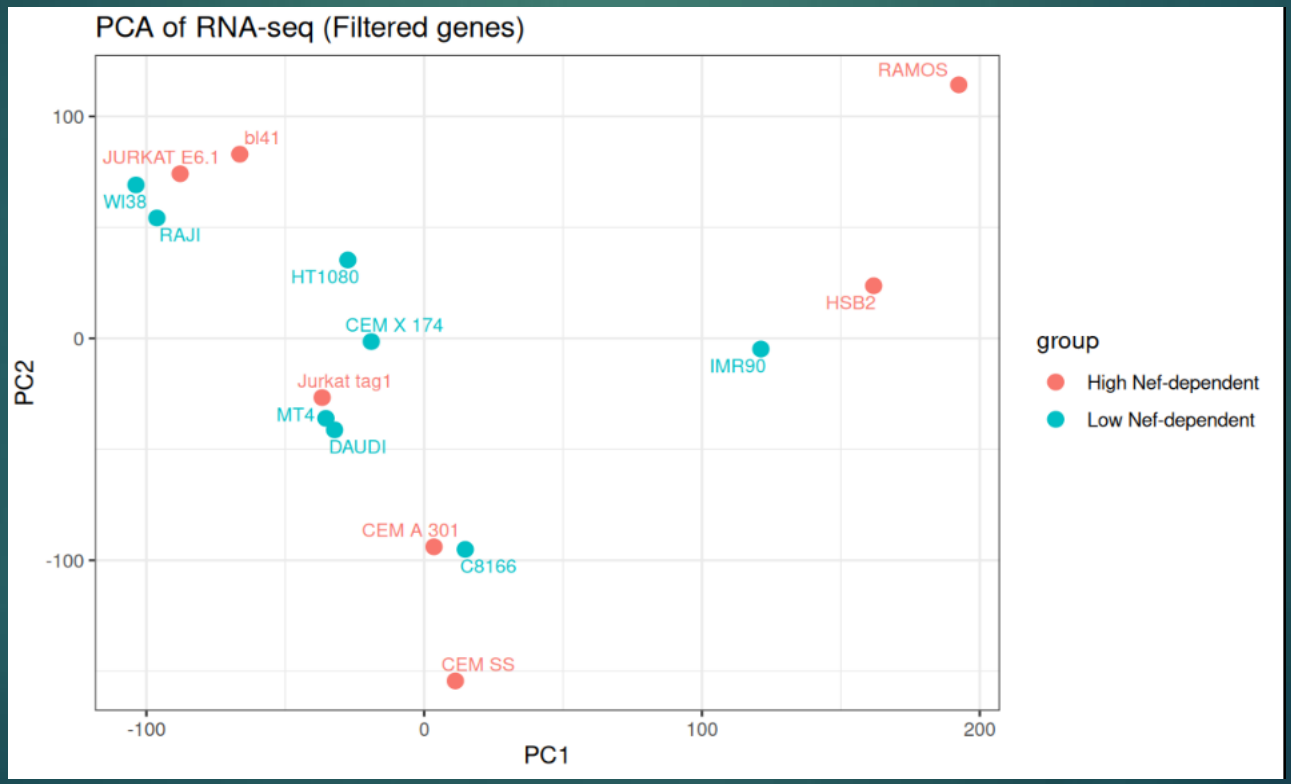
Spearman correlation:
rho = -0.01428571
p-value = 0.9642215

Pearson correlation:
r = -0.003577079
p-value = 0.9899055

Kendall correlation:
tau = -0.00952381
p-value = 1

H4: PCA Analysis

- Dimensionality reduction of RNA-seq data
- PC1/PC2 explain major variance
- Explores global expression structure



H5: SERINC5/SERINC3 Balance and Infectivity

Hypothesis

- The *relative expression balance* between SERINC5 and SERINC3 (SERINC5/SERINC3 ratio) predicts HIV-1 infectivity better than individual gene expression.

Rationale

- SERINC3 and SERINC5 are both restriction factors against HIV-1
- Previous results show:
  - Weak/absent individual correlation with infectivity
  - No strong co-expression between SERINC3 and SERINC5
- Therefore, a **functional balance model** may better capture biological effect than single-gene analysis

Analysis and Results

Statistical Tests Used

- Spearman correlation (rank-based, non-parametric)
- Pearson correlation (linear relationship check)
- Wilcoxon rank-sum test (High vs Low infectivity groups)

Results

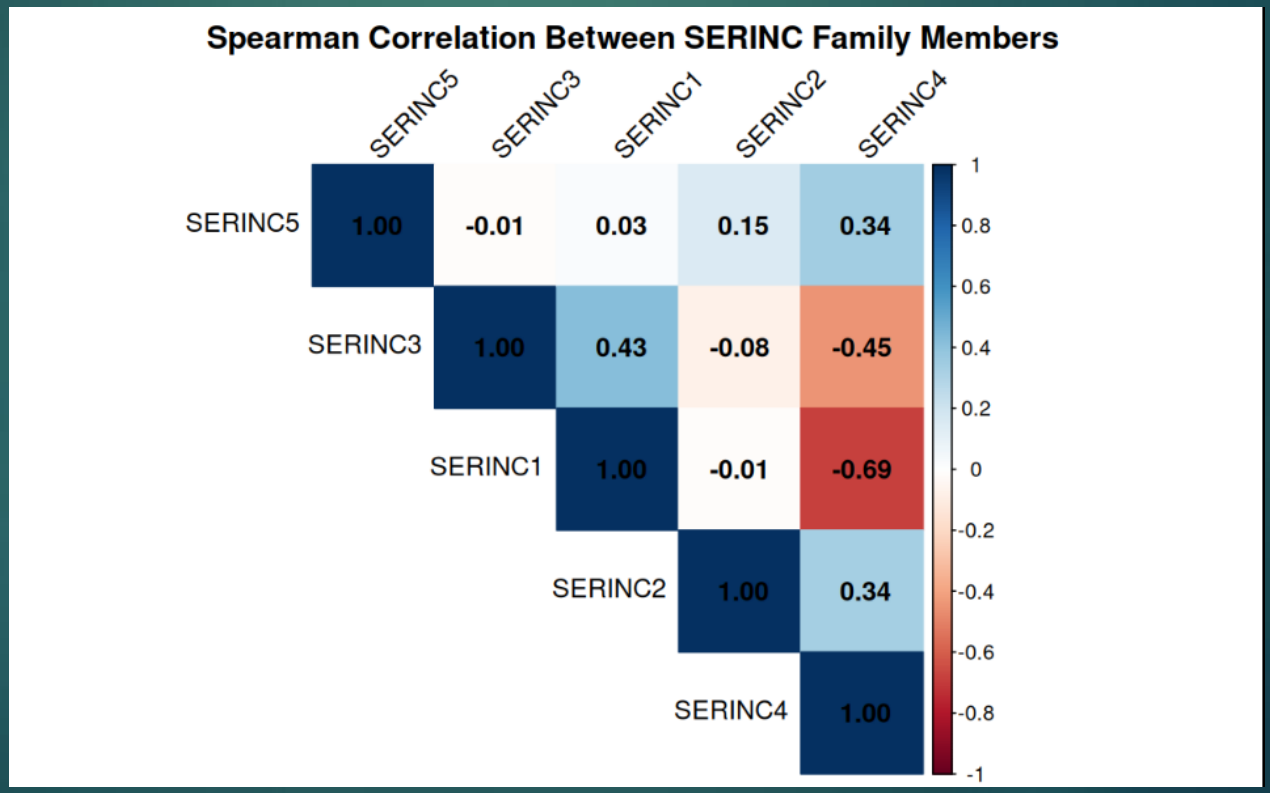
- Spearman: $\rho = -0.36$, $p = 0.187$
- Pearson: $r = -0.31$, $p = 0.264$
- Wilcoxon: $p = 0.336$
- Overall trend: **negative association (not statistically significant)**

- **Interpretation**
- Higher SERINC5 relative to SERINC3 shows a **weak tendency toward lower infectivity**
- However, the effect is **not statistically significant**
- Suggests infectivity is **not driven by SERINC balance alone**

- **Key Insight**
- SERINC genes likely act within a **broader regulatory network**, not as isolated predictors of HIV infectivity

SERINC Family Heatmap

- • Correlation between SERINC genes
- • Spearman correlation matrix



Key Findings

- • SERINC5 shows strongest association with infectivity
- • SERINC3 weaker / inconsistent association
- • No strong co-expression signal
- • PCA shows global variation across cell lines

Thank You